

Portfolio: <https://dylan-jacobs.github.io> GitHub: <https://github.com/dylan-jacobs> LinkedIn: <https://linkedin.com/in/dylan-t-jacobs/>

Education

Swarthmore College, Philadelphia, PA

Aug 2023—May 2027

Bachelors of Science in General Engineering & Bachelors of Arts in Applied Mathematics

Electrical Engineering **Teaching Assistant** for Electrical Circuit Analysis (ENGR 011)

Aug 2025—present

Relevant coursework:

- Electronic circuit analysis & applications, Digital signal processing, Data structures & algorithms in C++
- Ordinary & Partial differential equations (PDEs), Numerical PDEs, Real Analysis, Tensor decompositions
- Thermo-fluid mechanics, Electromagnetism, waves and optics with biomedical applications.

Overall GPA: 3.98/4.00

Research and work experience

Advanced Materials Intern & Research Assistant – US Naval Research Laboratory, Washington, DC May 2025-Aug 2025

- Measured thermal diffusivity and specific heat of thermoelectrics and semiconductors using laser flash analysis (LFA).
- Built Arduino-controlled LN₂ dispenser with closed-loop thermocouple monitoring, enabling automated LFA experiments
- Created graphical-user-interface and data-logging system interfacing between LN₂ dispenser and lab computer.
- Developed LFA and Seebeck analysis experiments to improve the lab's measurement procedure for near-IR-transparent and high-resistance samples; wrote standard operating procedures for the LN₂ dispenser and LFA.
- Taught myself fundamentals of nanomaterials science & solid-state physics to better understand the semiconductor growth process, the applications of my experiments, and to better communicate and present research updates to NRL scientists and engineers.

Applied Mathematics Research Assistant – Swarthmore College Mathematics, Philadelphia, PA

Jan 2024-present

- Utilizing principles of computational fluid dynamics and numerical methods to research high-order accurate methods for partial differential equations (PDEs), plasma/kinetic models in MATLAB.
- Developing low-rank, structure-preserving integrator for Vlasov-Fokker-Planck plasma equation in cylindrical coordinates.
- Poster presentations: 2025 SIAM NNP conference, 2024 & 2025 *Sigma Xi* Swarthmore poster sessions.

Electrical Engineering Research Assistant – Swarthmore College Engineering, Philadelphia, PA

Nov 2023—May 2024

- Researched electrical/aerospace technology behind wind-energy devices to develop oscillatory wind-energy harvester.
- Used Arduino, MATLAB & ViscousFlow to simulate air vortex-shedding & oscillatory electrical induction power output.

Software Engineering Summer Intern - Oregon Health and Science University, Portland, OR

Jun 2022—Aug 2022

- Used Kotlin to develop Android app that analyzes audio data from a Bluetooth stethoscope to detect heart murmurs.
- Presented machine-learning paper to the lab's reading group.
- Attended and presented weekly project updates and machine learning meetings

Projects

Analog Circuit and PCB Design: used Fusion360 to design and fabricate printed circuit boards (PCBs) Aug 2025—Dec 2025

- Combined MOSFETs, sensors, actuators, op-amps, comparators, audio amplifiers, and other integrated circuits
- Projects included [audio equalizers](#), [temp-controlled fans](#), [light-optimizing solar panels](#), & regulated DC power supplies.

AI Python Stock Trading: trained neural network on technical indicators to trade stocks using Alpaca API, [link](#) Feb 2025

Generative Adversarial Network (GAN): Image-generating model trained on web-scraped art, [link](#) Apr 2022—Feb 2023

FireSale: Used Java, AWS backend to build Android app to simultaneously reduce food waste and hunger, [link](#) 2020—2021

Technical & Professional Skills

Technical: SPICE, PCB design, CAD, Fusion360, embedded systems, PDE solvers, numerical methods

Programming: Python, MATLAB, Java, C++, HTML/JavaScript, LaTeX

Software: VSCode, MATLAB, Arduino, Git, SolidWorks, AutoCAD, MS Office

Languages: Spanish (Fluent), Global Seal of Bilinguality (2022), **Citizenship:** USA, Ireland

Swarthmore Varsity Men's Soccer

Aug 2023—present

Electrical Engineering Teaching Assistant: (Electrical Circuit Analysis ENGR 011)

Aug 2025—present

Awards and Scholarships

Delaware Valley Engineers Undergraduate Scholarship, Delaware Valley Engineers Society

Feb 2025

Donna Prentice Memorial Scholarship, American Society of Civil Engineers

Feb 2024

National Merit Scholarship, National Merit Scholarship Corporation

Apr 2023